Energy for Sustainable Development 88 (2025) 101794

Contents lists available at ScienceDirect

Energy for Sustainable Development

journal homepage: www.journals.elsevier.com/energy-for-sustainable-development

Improving vaccination access through off-grid electrification: An assessment in Ghana

Andrés Andrade Velásquez [a], Abdulrahman Alsanad [b], Katrin Lammers [a], Avia Linke [a,*], Nathan Johnson [b], Elena van Hove [b], Catherina Cader [a]

[a] Reiner Lemoine Institut, Rudower Chaussee 12D, 12489 Berlin, Germany
[b] Arizona State University, 1151 S Forest Ave, Tempe, AZ, USA

ARTICLE INFO

Keywords:
Energy access
Public health
GIS analysis
Multi-criteria analysis
Energy system modelling

ABSTRACT

Access to reliable, clean energy is essential for quality healthcare, particularly during global health crises like COVID-19. While electricity access in Ghana has improved, a significant urban–rural divide leaves many rural communities without a stable power supply, limiting their access to quality healthcare. Although electrification and healthcare planning are inherently connected, they are often approached separately.

This study develops a stepwise methodology to integrate both sectors by combining open-source spatial analysis, energy system modelling, and multi-criteria analysis. Using Ghana as a case study, it quantifies the population with restricted healthcare access and analyses electricity demand, including additional needs for COVID-19 care. Based on this, electrification strategies are formulated for underserved health care facilities, along with a prioritisation framework.

Findings reveal that seven percent of Ghana's health care facilities are in weak- or off-grid areas, including 172 public facilities responsible for vaccine distribution, which require reliable electricity to maintain cold chains. Electrifying these health care facilities could improve healthcare services for nearly one million people. Additionally, specialised COVID-19 treatment could increase electricity demand by up to 60 percent, leading to higher energy costs. The open-source approach of this study allows for adaptation to other country contexts, providing a scalable framework for healthcare electrification planning.

Introduction

Motivation

The COVID-19 pandemic exposed vulnerabilities in global health systems (Liu, Lee, & Lee, 2020), including in Ghana (Ofori et al., 2022). It highlighted the vital role of electricity in healthcare, especially for intensive care and vaccine cold chains. Vaccine sensitivity to temperature requires reliable electricity throughout distribution (Pursell, 2015). This also applies to vaccines for other diseases, such as hepatitis and rabies, and essential medicines like insulin. Intensive care units rely on electricity for respirators, oxygen concentrators, and other life-saving equipment (Homer Energy, 2023). Beyond COVID-19, access to reliable electricity is crucial for quality healthcare e.g. in emergency and maternity wards. Many developing countries face pandemic-exacerbated challenges, including weak supply chains, financial barriers, understaffing, and long distances to healthcare facilities (Geldsetzer et al., 2020).

At a time when vaccines were key to controlling COVID-19, sub-Saharan Africa faced severe immunisation challenges due to broken cold chains and staffing shortages (Bangura, Xiao, Qiu, Ouyang, & Chen, 2020). In Uganda, some facilities stored vaccines in neighbouring centres due to faulty fridges (Malande et al., 2019). A long-standing lack of reliable electricity, worsened by population growth outpacing rural electrification, limits essential health services (Rutherford, Mulholland, & Hill, 2010; Umeh, 2018).

Reliable electricity enables wider healthcare services and proper equipment use (Perera, Boyd, Wilkins, & Itty, 2015). It enables basic services like lighting for deliveries, emergency care, refrigeration, sterilisation, and the use of essential devices (World Health Organization & World Bank, 2014). The increasing need for complex medical interventions, such as imaging for cancer detection, further raises electricity demand. In Nepal, electrified areas retain more health workers than non-electrified ones (Perera et al., 2015). Improving reliable access to electricity in HCFs thus brings a range of rural development benefits,

* Corresponding author.
E-mail address: avia.linke@rl-institut.de (A. Linke).

https://doi.org/10.1016/j.esd.2025.101794
Received 18 November 2024; Received in revised form 15 July 2025; Accepted 21 July 2025
Available online 19 August 2025

demonstrating the importance of linking electrification and health planning.

Problem statement and approach

Access to reliable energy and quality healthcare are interlinked but rarely planned together. This study addresses the lack of planning methodologies for the energy-healthcare nexus by developing a replicable approach. We use geo-referenced data on health infrastructure, settlements, and night light imagery, focusing on Ghana to identify HCFs in weak or off-grid areas most vulnerable to electricity shortages and analyse their surrounding populations and combine this information with energy system modelling approaches. Using COVID-19 as a case study, we examine vaccine refrigeration and intensive care. Both require reliable electricity to power essential equipment. By assessing current treatment capacity and increased pandemic-related energy demand, we model cost-effective renewable mini-grid solutions for different HCF types. We then develop a prioritisation framework based on economic, environmental, and social criteria. Thus, this study integrates geospatial analysis, energy system modelling, and MCA to bridge the traditionally separate fields of electrification and healthcare planning — an approach not previously explored in the literature. It contributes a comprehensive approach for our case study Ghana, and can be applied to other country contexts, as it relies on open-source data and tools.

Aim and research questions

This study aims to develop an integrated approach to electrification and healthcare planning. We identify affected HCFs and populations, assess COVID-19 electricity needs, and model optimal energy systems for different HCF types considering their healthcare services they usually provide.

By combining the locations and types of HCFs in weak or off-grid areas, the HCFs's potential outreach (catchment area) with the mini-grid sizing, we are then able to make recommendations to prioritise the electrification according to specific criteria within a MCA. This helps local actors prioritise, since full electrification is unrealistic.

Our research addresses the following questions:

- Q1: How many HCFs lack access to reliable electricity and where are they located?
- Q2: What is the size of the population receiving limited health services due to poor access to electricity at selected facilities, and where are they located?
- Q3: What is the electricity demand impact if HCFs offer additional special care units for COVID-19 patients?
- Q4: What power generation sources best electrify HCFs in normal circumstances and under the COVID-19 pandemic case?
- Q5: What are the factors involved in prioritising the electrification of HCFs and how can we combine these factors to derive a prioritisation methodology?

Background

Healthcare system in Ghana

Ghana's healthcare system includes private and public HCFs. Private HCFs are owned and operated by individuals or organisations and are primarily funded through medical service payments (Agyemang-Duah, Peprah, & Peprah, 2020). A significant portion of these facilities falls under the private not-for-profit category, primarily run by mission and faith-based organisations (World Bank, 2011). Public HCFs are managed by the government through the Ministry of Health (MoH), delivering the majority of health services in Ghana (Agyemang-Duah et al., 2020; Dzando et al., 2022).

The National Health Insurance Scheme (NHIS), introduced between 2003 and 2005, aimed to improve healthcare accessibility and effectiveness (Dixon, Tenkorang, & Luginaah, 2013). As of Ghana's 2021 Census, nearly 70% of the population is covered by NHIS or private health insurance schemes (Ghana Statistical Service, 2021). Despite its success in expanding access, NHIS faces several challenges, particularly operating at a deficit (Umeh, 2018).

The public health system has three levels: primary, secondary, and tertiary (Fig. 1). Each level has defined personnel, service, and equipment requirements. Primary-level facilities, such as CHPSs, provide preventive care, maternal and child health services, and basic diagnostics (Asamani et al., 2018; Dzando et al., 2022). Required equipment includes diagnostic sets, sterilisers, thermometers, glucometers, and hospital beds (Health Facility Regulatory Agency (HeFRA), 2017d). Other primary facilities like health centres and polyclinics offer additional services such as minor surgeries and laboratory tests (Health Facility Regulatory Agency (HeFRA), 2017a, 2017e).

Secondary-level facilities include district and regional hospitals, while tertiary-level facilities comprise teaching hospitals (Asamani et al., 2018; Dzando et al., 2022). These facilities must be equipped with advanced medical devices such as ultrasound, X-ray, MRI, EEG, ECG, and CT scanners (Health Facility Regulatory Agency (HeFRA), 2017b, 2017c). Tertiary hospitals also provide specialised services, including ophthalmology, otorhinolaryngology, dental, and mental healthcare (Health Facility Regulatory Agency (HeFRA), 2017b, 2017c).

Higher-tier HCFs demand more electricity due to their advanced equipment and specialised services. However, many HCFs in off-grid or weak-grid areas experience power instability, hindering service delivery. This persistent energy challenge remains a critical barrier to effective healthcare delivery in developing countries (World Health Organization & World Bank, 2014).

Therefore, the following section aims to shed light on the link between electricity and healthcare in the Ghanaian context.

Electrification status in Ghana and its impact on healthcare

According to the International Energy Agency and the World Bank (IEA, IRENA, UNSD, World Bank, and WHO, 2021; World Bank, 2020a), Ghana's electricity access rose from 42.6% in 1998 to 83.5% in 2019. Despite this progress, a significant urban-rural divide persists: as of 2020, approximately 95% of urban areas had electricity, compared to 74% of rural areas (World Bank, 2020b, 2020c). Many rural settlements remain in weak- or off-grid areas (IEA, IRENA, UNSD, World Bank, and WHO, 2021). Even grid-connected regions face frequent power outages, transmission losses of up to 25%, and voltage fluctuations that damage sensitive medical equipment (Nyarko Kumi, 2020; Sakah, Diawuo, Katzenbach, & Gyamfi, 2017).

Modern energy is crucial for resilient healthcare (Keim, Pastukhova, Voss, & Westphal, 2019). Research has shown that reliable energy access significantly impacts healthcare delivery in developing countries, particularly in rural areas where facilities struggle with maintaining essential services during power disruptions (Abdulkarim et al., 2024). A comprehensive study examining energy challenges in healthcare facilities found that unreliable power supply affects critical services including vaccine refrigeration, emergency care, and basic medical procedures (Khogali et al., 2022). Moreover, properly electrified HCFs could attract and maintain skilled health workers, helping combat personnel shortages in rural areas (World Health Organization & World Bank, 2014).

Recent studies have demonstrated that energy poverty significantly impacts healthcare delivery in Ghana and similar regions. Research conducted in West Africa revealed that weak grid connections and unreliable power supply compromise the effectiveness of HCFs, particularly affecting rural communities (Khogali et al., 2022; Ngonyani & Arthur, 2024; Shastry & Rai, 2021). The COVID-19 pandemic further exposed these vulnerabilities, straining under-resourced HCFs

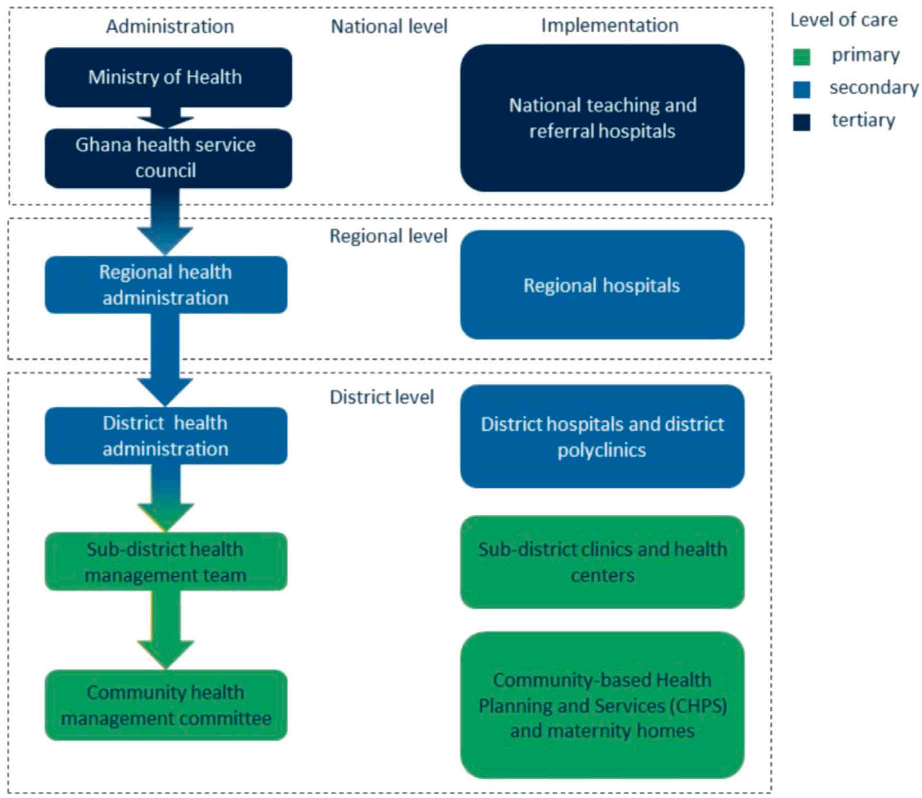


Fig. 1. Institutional Structure of Ghana's Healthcare System (own visualisation, based on UNF, ASD, and Ministry of Power Ghana, 2015).

(Afulani et al., 2021; Lazo, Escobar, & Watts, 2023). Using geospatial analysis, this study visualises the location of weak- and off-grid HCFs and estimates how many surrounding households and settlements would gain improved healthcare access through their electrification.

Recognising this energy-health link, the study supports sustainable energy transition in Ghana's health sector. It develops a multi-methodological approach based on open-access, open-source tools and data, ensuring its applicability beyond the case study to other countries and regions. In the following chapters, the reader will be guided through the applied methods and the results of this study, which will be carried out in the following steps:

- Allocation and visualisation of HCFs located in weak or off-grid areas, including their surrounding communities within walking distance,
- Determination of electricity demand of different HCFs types taking into account additional devices for treatment and prevention of COVID-19,
- Energy system simulation for assessed HCFs types,
- Application of a multi-criteria decision framework to develop a prioritisation strategy for HCFs electrification.

Methodology

This section outlines the study's methods (Fig. 2): geospatial analysis of Ghana's HCFs (grey), energy demand and system modelling under COVID-19 (green), and MCA for prioritising power generation and electrification based on economic, environmental, and spatial factors (blue).

Geospatial analysis

Available datasets

Healthcare facilities: Georeferenced HCF data from Ghana Open Data Initiative (GODI) (2020) included 2614 cleaned entries (Ghana Open Data Initiative, 2020). A heat density map revealed concentration in southern urban areas and sparse distribution in northern regions (Fig. 3).

Population: Population data combined high-resolution population density maps from Center for International Earth Science Information Network (CIESIN) (2016) and Geo-Referenced Infrastructure and Demographic Data for Development (GRID3) (2020) sources (Center for International Earth Science Information Network (CIESIN), Columbia University and Novel-T, 2020; Facebook Connectivity Lab and Center for International Earth Science Information Network - CIESIN - Columbia University, 2016). Settlements were categorised into Built Up Areas (BUAs), Small Settlement Areas (SSAs), and hamlets based on size and building density. Population data was integrated with settlement polygons, applying buffers to account for recent growth (Fig. 4).

Energy Infrastructure: Electrical grid network data from the Global Predictive Mapping Project (2020) and 16-bit raster night lights data from NASA's Black Marble nighttime lights product, which belongs to the Visible Infrared Imaging Radiometer Suite (VIIRS) (2018) were utilised (Arderne, Zorn, Nicolas, & Koks, 2020; Román et al., 2018). Night light data indicate electrification patterns but have limitations, such as 450 m resolution and urban bias that may overstate urban and understate rural electrification.

Data processing

Categorisation of electrified and non-electrified healthcare facilities: Electrified and non-electrified HCFs were categorised using VIIRS night lights data (450 m resolution). Night-light data were thresholded (digital numbers (DNs) = 0–100), then converted to 8-bit (DNs = 0–255) for clarity. A threshold of DN = 50 was set to separate electrified/non-electrified areas based on cross-referencing with daylight images. This threshold value was used to create a binary mask of potentially electrified vs. non-electrified areas for all of Ghana (Fig. 5).

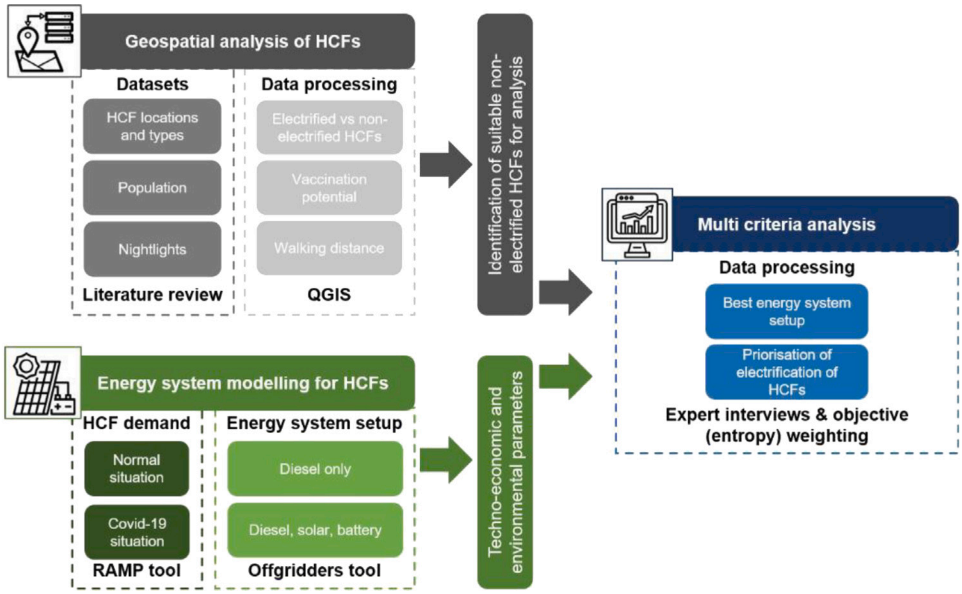


Fig. 2. Overview of applied methodologies and consecutive steps.

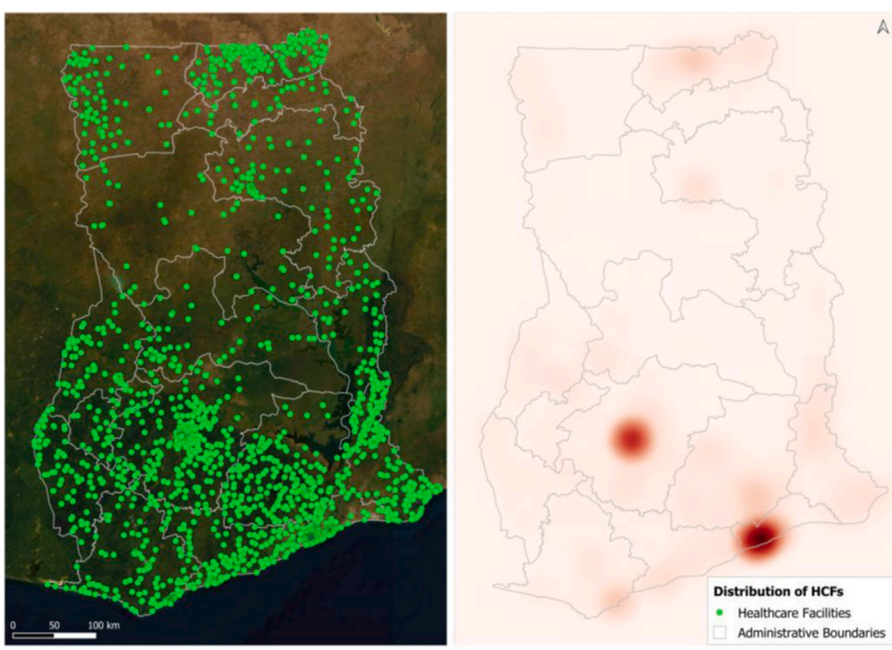


Fig. 3. Total HCF dataset (left) and heat density map (right).

However, VIIRS data use poses challenges, particularly in rural and sparsely populated areas where faint or dispersed lighting sources may fall below the detection threshold. This can lead to a potential misclassification of electrified facilities in areas where lighting is too weak to be detected. Conversely, bright reflections from urban centres may cause overestimation of electrification. To mitigate these effects, we cross-referenced the VIIRS Black Marble data with the population settlements data and daylight imagery.

Finally, the QGIS NN Join plug-in was used to determine distances between HCFs and night light sources/electrified zones.

Categorisation of healthcare facilities with and without vaccination distribution potential: HCFs with vaccination distribution potential were identified based on COVID vaccine availability in public HCFs during Ghana's 2022 campaign (Ghana Health Services, 2022). Official (public) health centres, including primary HCFs, hospitals, long-term care facilities, community care centres, and day care centres, were considered as sites with vaccine distribution potential. The dataset was divided into electrified/non-electrified facilities and those with/without vaccination potential. This categorisation allowed for the identification of non-electrified facilities with vaccination potential, while excluding electrified facilities. The decision to consider all health sites assumed to be non-electrified in the energy system modelling and the MCA, regardless of their current vaccination potential, was based on ethical considerations and the potential for non-electrified facilities without vaccination potential to be used for emergency vaccination campaigns if properly electrified. Therefore, four types of HCFs were considered relevant for this analysis, as they are located in off-grid and weak grid areas: CHPS, Health Centre, Maternity Home & RCH, and

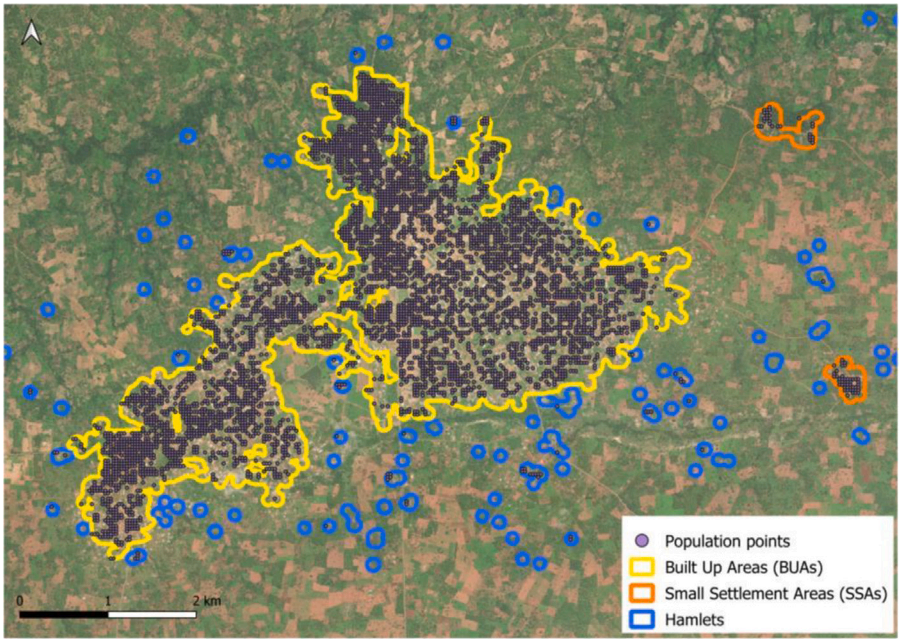


Fig. 4. Population points within settlement polygons.

Clinic (according to their Health Facility Regulatory Agenda (HeFRA) categorisation).

Determination and application of maximum walking distance: Walking was chosen as the most accessible transport mode for Ghanaian households. A maximum distance of 5 km between a settlement and one of the selected HCFs was chosen, based on a survey by Paköz and Yüzer in Istanbul in 2014 (Paköz & Yüzer, 2014). We determined the areas covered by this 5 km radius using buffers in QGIS. However, we decided to explicitly choose only those settlements whose centroids lay within the maximum 5 km buffer. Thus, settlements that for example only bordered with the buffer were not considered for our analysis.

Demand and energy system modelling

The study determined the most suitable mini-grid setup for each site, serving as a key input for the MCA. The energy system was simulated and optimised for each HCFs site based on various input criteria, including energy demand, investment cost for different energy system components, local fuel prices, and cost of capital. While most parameters were obtained through literature review, the energy demand was estimated and modelled, with a focus on the additional energy demand required to operate COVID-19 treatment wards.

Demand modelling

The demand modelling process involved estimating the energy demand at each site using the stochastic energy demand modelling tool RAMP (Lombardi, Balderrama, Quoilin, & Colombo, 2019). This tool generates detailed daily, weekly, or yearly load profiles at minutely and hourly resolutions based on appliance inventories and user behaviour patterns. RAMP is particularly valuable in the absence of metered data, as it provides user-driven energy demand time series, which was essential for our analysis (Lombardi et al., 2024). Widely applied in research projects across various country contexts, RAMP has been used for GIS-based rural electrification planning in Ethiopia (Dimovski et al., 2024), renewable energy (RE) system planning in rural Vietnam using survey data (Hart, Eckhoff, & Breitner, 2024), and the development of demand archetypes for rural users in Sub-Saharan Africa (Stevanato, Sangiorgio, Mereu, & Colombo, 2023). As one of the few open-source tools capable of generating sophisticated load profiles from basic input data – such as appliance lists – it has proven highly useful for rural electrification planning and was therefore used in our study. The equipment used in each site was determined by the four different HCF types considered in this analysis (CHPS, Health centre, Maternity Home & RCH, and Clinic) and their usual healthcare services and associated equipment requirements according to HeFRA (Health Facilities Regulatory Agency, 2023). The resulting appliance list for each HCF type was one of the inputs for the RAMP tool. The other important factor to model the demand was the user behaviour and power ratings, which were provided by the HOMER Powering Health tool (Homer Energy, 2023) for different HCF types. Service levels and appliances vary across the four HCFs types; demand was assessed separately for all four HCF types, resulting in four different typical annual load profiles respectively. In addition, load profiles were determined for both a normal situation and a COVID-19 pandemic situation, with a focus on the additional energy demand required to operate COVID-19 treatment wards. By using RAMP, stochastic variability is applied, and therefore, daily load profiles will differ slightly from each other. The tool also takes into account differences between weekday and weekend loads based on the time of use inputs entered and applied for this study. Seasonal demand variation was excluded due to limited data and the nature of the HCF types analysed, which mostly operate all year round without air conditioning.

Energy system modelling

The energy system modelling tool Offgridders, developed by Reiner Lemoine Institut (RLI) (Hoffmann, 2019a), is a Python-based tool used internationally to optimise and simulate off- and on-grid mini-grid energy systems, and to run sensitivity analyses to evaluate feasible energy system configurations. Offgridders stands out from other tools due to its ability to optimise and simulate multiple sites at once, making it particularly suitable for this study. The tool and its results have been compared with other common modelling tools (Hoffmann, 2019b) and applied to case studies such as micro-grid planning with different types of battery storage in Morocco (Mahir et al., 2024) or multi-level optimisation of rural electrification planning at regional level in Nigeria (Juanpera, Ferrer-Martí, & Pastor, 2022).

Input parameters included component-specific inputs, economic parameters, general characteristics, RE potential, and demand profiles for each HCF. Demand profiles were determined using the RAMP tool

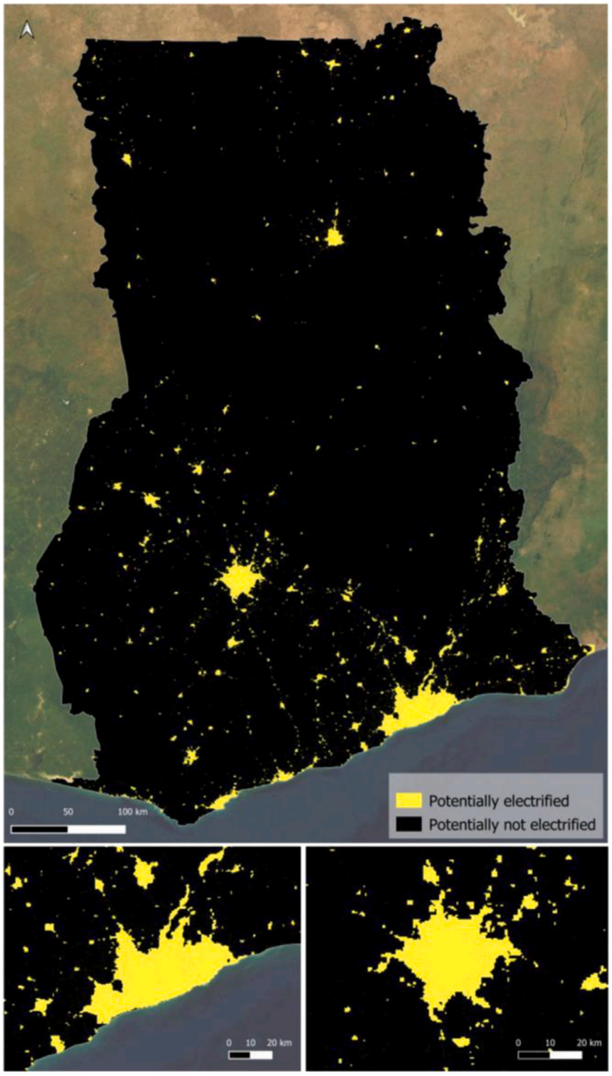


Fig. 5. Binary night lights image from Ghana. Areas assumed to be electrified are depicted in yellow. Close ups to the cities of Accra (bottom left) and Kumasi (bottom right).

as explained above. Simulations generated economic, technical, and environmental outputs, indicating the most suitable energy system setup for each site. Two system setups were analysed: a diesel-only system and a photovoltaic (PV)-battery-diesel system.

Outputs included LCOE, optimal component sizes, investment costs, operation and maintenance costs, diesel fuel consumption, and CO_2 emissions. These results are crucial for understanding how the COVID-19 pandemic affects energy demand and techno-economic parameters of energy systems supplying HCFs with COVID-19 wards implemented. They form the basis for creating the MCA in subsequent chapters, providing most criteria relevant for this procedure.

Multi-criteria analysis

The MCA is a valuable tool for complex decision-making in energy planning and system simulations. This study employed the Compromise Ranking method (VIKOR) to gauge proximity to the "ideal" solution and generate rankings (Opricovic & Tzeng, 2004; San Cristóbal, 2011).

Overview of multi-criteria analysis for sustainable energy problems

The MCA involves: defining alternatives and criteria, weighting, evaluation, and ranking. Criteria selection typically encompasses technical, economic, environmental, and social aspects (Juanpera, Blechinger, Ferrer-Martí, Hoffmann, & Pastor, 2020; Wang, Jing, Zhang, & Zhao, 2009).

Criteria Selection: Relevant criteria were selected based on principles like systematic representation of energy system performance, alignment with MCA objectives, criterion independence, measurability, and comparability (Si, Marjanovic-Halburd, Nasiri, & Bell, 2016; Wang et al., 2009). Common criteria in energy-related decision-making include efficiency, investment costs, CO_2 emissions, and job creation (Wang et al., 2009).

Criteria Weighting: Two methods were used for criteria weighting: subjective weighting based on expert input and objective weighting using the entropy method (Deng, Yeh, & Willis, 2000). Subjective weighting was employed because expert judgment provides critical qualitative insights, particularly for social and policy-related criteria that may lack robust quantitative data. However, recognising the potential for bias, a structured questionnaire was distributed to 12 experts with various relevant backgrounds in energy systems and healthcare infrastructure. Experts rated each criterion's importance on a five-point Likert scale (1 = low, 5 = high).

The criteria included economic, environmental, and geospatial dimensions and varied depending on the specific multi-criteria analysis. For each set of criteria, the individual importance scores were aggregated by summing the responses for each criterion across all experts. The resulting totals were then normalised by dividing each criterion's total score by the sum of all scores within that analysis. The normalised values were used as the final weights in the multi-criteria evaluation.

To complement subjective weighting, the entropy method was applied as an objective weighting technique. This method assigns weights based on the inherent variability of data, reducing dependence on subjective judgment. It was chosen over alternative objective methods (e.g. Criteria Importance Through Intercriteria Correlation (CRITIC)) due to its ability to handle diverse criteria scales and emphasise criteria with higher discrimination power (Deng et al., 2000).

Other weighting approaches, such as Analytic Hierarchy Process (AHP) or equal weighting, were considered but not selected. AHP, while effective, can introduce inconsistencies in pairwise comparisons, particularly when handling many criteria (Ishizaka & Labib, 2011). Equal weighting, though simple, does not account for differences in criteria importance.

By combining subjective and objective methods, this approach balances expert insights with data-driven decision-making, enhancing the robustness of the MCA process.

Ranking According to the VIKOR Method: The VIKOR method compares each alternative to an ideal solution that performs optimally for all criteria (Juanpera et al., 2020). Alternatives were ranked by distance to the ideal solution (Opricovic & Tzeng, 2004).

Application of multi-criteria analysis to case study

Two MCAs were conducted: First, evaluating the best energy system setup per site. Followed by prioritising electrification of HCFs. Both MCAs followed similar steps: alternative definition, criteria definition, criteria weighting, and alternatives evaluation and ranking using the VIKOR method.

For the first MCA, alternatives were diesel-only, PV-diesel, and storage mini-grids. Criteria included initial investment costs, O&M costs, LCOE, and CO_2 emissions.

The second MCA prioritised HCF electrification based on economic and spatial criteria, including initial investment costs, LCOE, number of households reached, and distance to the nearest grid connection point.

MCAs were implemented using Python 3.9.13 in a Jupyter notebook environment, utilising pandas and matplotlib libraries (Papathanasiou & Ploskas, 2018).

Results

This chapter presents results from the applied methods, structured into three parts: geospatial analysis, energy system modelling, and the MCA.

Geospatial analysis

Categorisation of healthcare facilities

Our processed HCFs dataset yielded 2614 points—somewhat lower than literature estimates of 3200–3500 public and private facilities in Ghana from 2014–2017 (Jehu-Appiah et al., 2014; Wang, Otoo, & Dsane-Selby, 2017). This difference stems from refined processing methods that excluded mislocated facilities, such as those in water bodies or deserted areas. By ensuring only valid data points were included, we aimed for a more accurate representation of HCFs in the country. This approach aligns with similar studies, such as those by Jehu-Appiah et al. who also faced challenges in data accuracy when mapping facilities in rural regions (Jehu-Appiah et al., 2014).

From our analysis, we found that the vast majority of HCFs (73.9%) were located within or no further than 1.1 km from an electrified zone (see Section "Data Processing"). Furthermore, 93.9% of HCFs lay within 9.9 to 11 km of an electrified area. This finding is consistent with previous studies, such as those by Wang et al. which highlighted similar distances in their analysis of electrification gaps in sub-Saharan Africa (Wang et al., 2017).

We applied a 10 km threshold for selecting HCFs that could potentially be served by PV mini-grids instead of expanding the national grid. Of the 2614 HCFs across Ghana, 184 (7.04%) were located beyond this threshold, suggesting they are either non-electrified or situated in weak- or off-grid areas that are more suited to off-grid electrification solutions (Fig. 6(a)). This is in line with findings from other studies, which showed similar distribution patterns of HCFs in off-grid regions of Africa (Florio, Freire, & Melchiorri, 2023). From these, 172 are public centres and thus viable vaccine distribution sites, according to Ghana's MoH (Fig. 6(b)).

Additionally, we categorised these 184 HCFs into sub-categories based on their electrification and vaccination potential status (Fig. 7). The chart highlights the number of HCFs that were either electrified or not, with and without vaccination potential. This categorisation is crucial for understanding how energy access intersects with healthcare provision, as non-electrified HCFs may face substantial challenges in maintaining quality services. Our study emphasises the importance of addressing both energy and healthcare needs, similar to the work by the World Health Organization (WHO) and others, which demonstrated the interdependence of electricity access and healthcare delivery in remote areas (Reuland et al., 2019; World Health Organization, The World Bank, and International Renewable Energy Agency, and Sustainable Energy for All, 2023).

Though 172 of the non-electrified HCFs were identified as most suitable for vaccine distribution, both the energy system modelling and the MCA in subsequent sections consider all 184 non-electrified HCFs. This decision is informed by ethical considerations and the recognition that any HCF lacking reliable electricity can benefit from a sustainable power supply, which could improve healthcare access for surrounding communities, as evidenced by studies in other African countries (Moner-Girona, Kakoulaki, Falchetta, Weiss, & Taylor, 2021; Reuland et al., 2019; World Health Organization, The World Bank, and International Renewable Energy Agency, and Sustainable Energy for All, 2023).

Analysis of population coverage

From our data, we estimated a total population of 30,405,027 for 2020 in Ghana. The World Bank's (World Bank, 2023) population estimation for the same year was 32,180,401. This discrepancy lies in the fact that the settlement extents were not able to capture all of the available population points in the point-in-polygon operation specified in Section "Available Datasets".

From our categorisation of HCFs into electrified and non-electrified, as well as into those with and without vaccination potential, we calculated the respective population coverage within a 5 km radius. Table 1 summarises these results.

However, we acknowledge that healthcare accessibility is influenced by factors beyond Euclidean distance, such as terrain, road networks, and mobility constraints (e.g., transport availability, socioeconomic barriers). While the 5 km threshold is a widely used benchmark in global healthcare access studies, it may oversimplify real-world conditions:

- In rural and mountainous areas, physical barriers (e.g., rivers, poor road conditions) may reduce actual accessibility, leading to an overestimation of coverage.
- Conversely, in urban and peri-urban areas, people may travel beyond 5 km to reach a well-equipped facility, meaning that our analysis could underestimate accessibility.

Despite these limitations, our results indicate that 85.7% of the population has at least partial healthcare access within 5 km (Fig. 8(a)). However, 4,340,778 people (14.3%) remain outside this radius, highlighting significant accessibility gaps (Fig. 8(b)). If the 172 HCFs with vaccination potential were electrified, approximately 933,514 people would gain improved healthcare access within walking distance. This number would increase to around 970,000 if all currently non-electrified HCFs were electrified (Table 1).

Demand and energy system modelling

After identifying all the HCFs located in off-grid and weak-grid areas, the energy demands of these HCFs were modelled to simulate and optimise suitable off-grid energy systems to supply the HCFs.

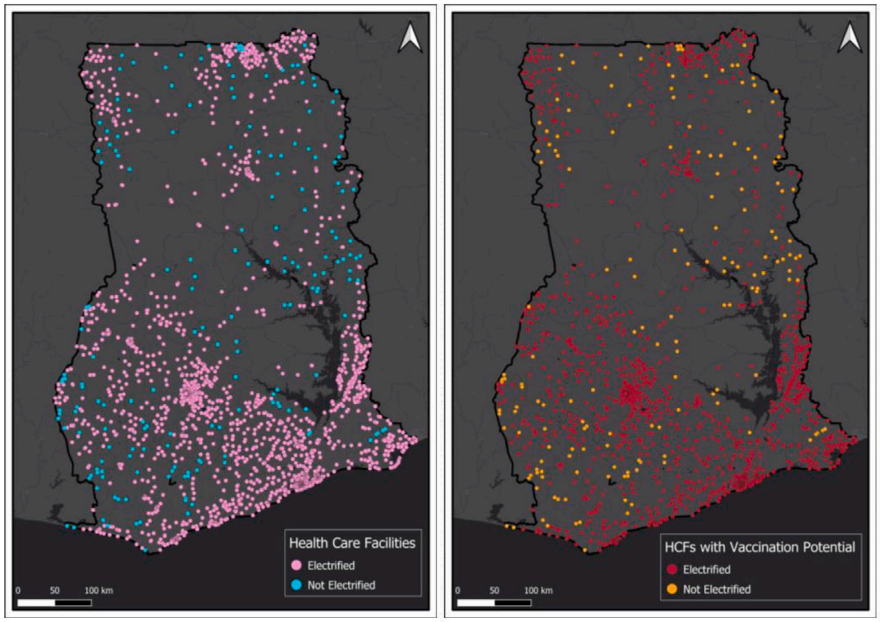
Demand modelling

The input parameters (appliance list including user behaviour) to run the RAMP tool as well as the respective output files (load profile for normal and pandemic situations) can be found in Appendix "Appendix_demand". According to HeFRA, of the four different types of HCF (CHPS, maternity home/RCH, health centre and clinic), only two types provide inpatient services: health centres and clinics. Therefore, COVID wards (as inpatient wards) and their additional energy demand (COVID appliances) compared to the normal situation were only considered for these two types of HCFs. Table 2 summarises the average and maximum energy demands for health centres and clinics in both normal and pandemic situations, along with the respective demands for maternity homes and CHPS in the normal situation (these two types usually had no intensive care units for COVID-19 patients) as calculated by RAMP.

The health centre and clinic demand were significantly higher when comparing the normal and pandemic situation:

- the average energy demand of health centres raised by 46%
- the average energy demand of clinics raised by 60%
- the maximum energy demand of both HCF types raised by 14%.

Fig. 9 shows a representative weekly load profile for both HCF types with COVID-19 treatment devices (clinic and health centre) for normal and COVID-19 pandemic situations. The pink profiles representing the COVID-19 situation are consistently higher than the curves representing the normal situation. Some peaks in the pink lines exceed those in the



(a) (b)

Fig. 6. Distinction between non- and electrified HCFs in Ghana, based on satellite nightlight imagery (a); Distinction between HCFs with or without vaccination potential (b)

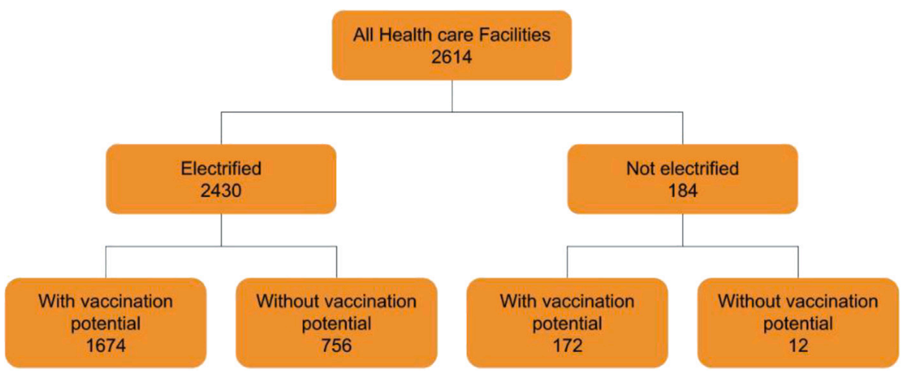


Fig. 7. Categorisation of HCFs according to their electrification and vaccination potential status.

Table 1
Population coverage according to health care facility categorisation.

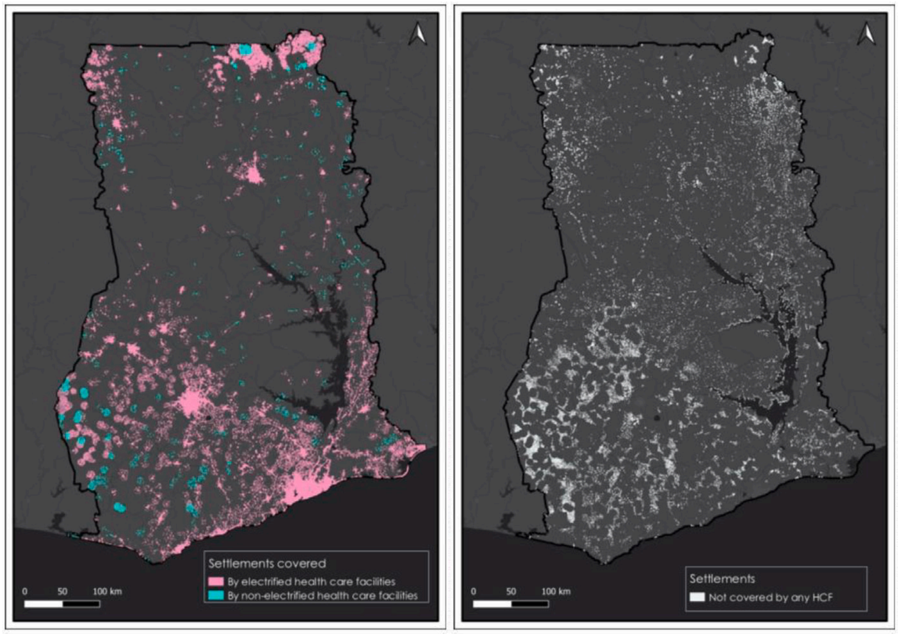
Health care facilities classes	Count	Coverage — Overall population 5 km buffer
All HCFs	2614	26,049,930
All HCFs with vaccination potential	1842	24,752,810
Electrified HCFs	2426	25,214,967
Electr. HCFs with vaccination potential	1670	23,891,000
Non electrified HCFs	184	969,083
Non electr. HCFs with vaccination potential	172	933,514

normal situation curves due to randomisation in the generation of the load profiles by RAMP.

In summary, only two types of the analysed HCFs offered specific COVID-19 treatments, clinics and health centres. Maternity homes and CHPSs usually have no facilities for inpatient services, which are required for COVID-19 patient treatments. Therefore, their energy demand did not increase. In contrast, the additional COVID-19 treatments and equipment required by clinics and health centres led to higher energy demand. Their percentage peak demand increased similarly, while the percentage increase in average energy demand was greater for clinics, the largest HCF type assessed. These findings are consistent with studies such as those by Jiang et al. which also found a significant increase in energy demand for HCFs during the pandemic due to the introduction of COVID-19-related treatment devices (Jiang, Klemeš, Fan, Fu, & Bee, 2021). In comparison to other regions, the magnitude of increase observed in the present study was higher for clinics, highlighting the particular strain faced by larger facilities.

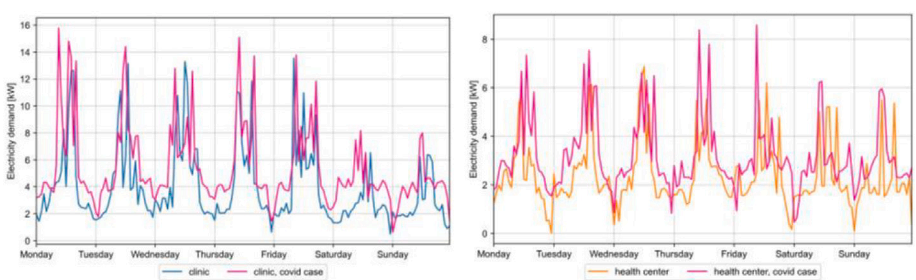
Energy system modelling

With the demand data for the four different HCFs, we were able to simulate and optimise their respective energy supply systems. In addition to the demand characteristics, techno-economic parameters were required to run these simulations with Offgridders, as described



(a) (b)

Fig. 8. Map of settlements in Ghana covered by electrified and non-electrified HCFs (a) and settlements not covered by any HCF (b).



(a) Load profile clinic

(b) Load profile health centre

Fig. 9. Representative weekly load profiles for the HCF type clinic (a) and health centre (b) for normal and COVID-19 pandemic situations.

Table 2
Energy demand of four HCF types comparing the normal and – if applicable – the pandemic situation based on a yearly load profile in hourly resolution created by the RAMP tool.

	Normal situation		COVID-19 situation	
	Average [kWh]	Maximum [kWh]	Average [kWh]	Maximum [kWh]
Health centre	2.37	7.68	3.18	8.58
Clinic	3.84	17.15	5.45	18.48
Maternity home	2.54	7.82	N/A	N/A
CHPS	1.85	6.96	N/A	N/A

in the Methodology section. A summary of these input parameters and their sources can be found in Appendix "Energy system modelling".

Table 3 shows the minimum and maximum LCOE values for all simulated and optimised 184 HCF sites. The LCOE provides a useful first impression of the economic viability of energy projects, as it accounts for all costs over the project's lifetime, including investment, operation and maintenance, and replacement costs. The LCOE is therefore a valuable parameter when evaluating different electrification options and sites.

The diesel setup had significantly higher LCOE and was therefore economically unfeasible compared to the PV-battery-diesel setup. While the maximum values were the same when comparing the normal and COVID-19 pandemic situations (due to higher demands), the minimum values showed lower results for the COVID-19 situation (0.04 USD/kWh reduction). The reason for this is that the maximum LCOE values were determined by the smaller HCF types CHPS, maternity homes and RCH, which had higher electricity costs, because their overall investment, operation and maintenance as well as replacement costs were distributed across a smaller share of electricity sales (consumption). Moreover, these types were not facing demand increases due to the COVID-19 pandemic as they do not typically provide COVID-19 treatment. Larger HCF types, like clinics and health centres, had higher demand and also faced increased electricity demand due to COVID-19-related appliances. As a result, overall costs were distributed over higher units of electricity being consumed, reducing the cost per kWh.

The following four tables show a more detailed overview of simulation results for all four HCF types. The values represent the average for each type, as slight differences in capacity and cost arose from variations in solar irradiation across the country.

In general, LCOEs were lower for the PV-battery-diesel system setup than for the 100% diesel setup. Even though the investment cost for the diesel setup was lower, maintenance and operation costs were significantly higher due to the constant need for diesel fuel supply, which led to higher LCOE. These results corroborate findings from previous studies, which suggest that hybrid systems, while more

Table 3
Overview of LCOE as a result of energy system simulation for 184 HCFs - Minimum and maximum values.

Energy system setup	Normal situation		COVID-19 situation	
	Min. [USD/kWh]	Max. [USD/kWh]	Min. [USD/kWh]	Max. [USD/kWh]
Solar, battery, diesel	0.30	0.45	0.26	0.45
Diesel	0.94	1.05	0.90	1.05

Table 4
Results of energy system modelling for HCF type clinic representing the normal and COVID-19 pandemic situation for two different energy system setups — average values of all 40 simulated cases and a project lifetime of 25 years.

Parameter	Solar, battery, diesel system		Diesel system	
	Normal	COVID-19	Normal	COVID-19
LCOE [USD/kWh]	0.33	0.29	0.95	0.91
Solar capacity [kW]	37.68	52.31	–	–
Battery capacity [kWh]	58.49	84.50	–	–
Diesel capacity [kW]	2.28	2.42	16.41	18.48
RE share [%]	0.98	0.98	–	–
CO_2 emissions [kg CO_2 equivalent]	832	1252	26,176	37,295
Investment cost [USD]	88,850	106,971	43,466	43,843
Operation & maintenance cost [USD]	7930	8347	33,059	44,084

Table 5
Results of energy system modelling for HCF type health centre representing the normal and COVID-19 pandemic situation for two different energy system setups — average values of all 83 simulated cases and a project lifetime of 25 years.

Parameter	Solar, battery, diesel system		Diesel system	
	Normal	COVID-19	Normal	COVID-19
LCOE [USD/kWh]	0.37	0.33	1.00	0.95
Solar capacity [kW]	21.50	28.81	–	–
Battery capacity [kWh]	38.61	53.50	–	–
Diesel capacity [kW]	1.11	1.49	7.71	8.50
RE share [%]	0.98	0.98	–	–
CO_2 emissions [kg CO_2 equivalent]	519	713	16,685	22,281
Investment cost [USD]	68,144	77,765	41,276	41,495
Operation & maintenance cost [USD]	7619	7812	23,649	29,197

Table 6
Results of energy system modelling for HCF type maternity home and RCH representing the normal and COVID-19 pandemic situation for two different energy system setups — average values of all 13 simulated cases and a project lifetime of 25 years.

Parameter	Solar, battery & diesel system	Diesel system
LCOE [USD/kWh]	0.37	0.98
Solar capacity [kW]	24.30	–
Battery capacity [kWh]	44.21	–
Diesel capacity [kW]	1.01	7.80
RE share [%]	0.98	–
CO_2 emissions [kg CO_2 equivalent]	496	17,946
Investment cost [USD]	71,799	41,315
Operation & maintenance cost [USD]	7597	24,899

Table 7
Results of energy system modelling for HCF type CHPS representing the normal and COVID-19 pandemic situation for two different energy system setups — average values of all 48 simulated cases and a project lifetime of 25 years.

Parameter	Solar, battery & diesel system	Diesel system
LCOE [USD/kWh]	0.42	1.04
Solar capacity [kW]	17.89	–
Battery capacity [kWh]	34.91	–
Diesel capacity [kW]	1.01	7.40
RE share [%]	0.98	–
CO_2 emissions [kg CO_2 equivalent]	441	14,071
Investment cost [USD]	64,302	41,215
Operation & maintenance cost [USD]	7542	21,057

capital-intensive upfront, provide greater long-term cost savings due to the reduced reliance on expensive diesel fuel (Omar, 2025).

Fig. 10 depicts the distribution of LCOE for all modelled HCF sites, across different system setups and scenarios. The blue and violet circles represent the diesel-only system setup, while the green and pink ones represent the hybrid energy system setup. The LCOE for the hybrid systems ranged from 0.26 USD/kWh to 0.45 USD/kW, whereas the diesel setups ranged from 0.89 USD/kWh up to 1.04 USD/kWh. This highlights that the diesel-only system setups had LCOEs two to three times higher than the PV-battery-diesel system setups.

Fig. 11 shows the initial investment required to install an energy system on 184 simulated sites. The first investment only accounts for the project and system component costs at the beginning of the project. The diesel system initial investments ranged from 41,100 USD to 43,908 USD, while hybrid system investments ranged

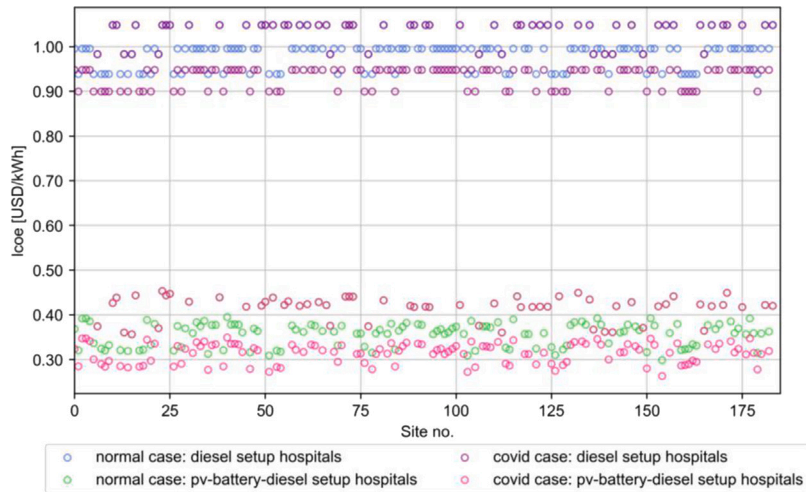


Fig. 10. ESM result: Levelised cost of electricity for different system setups under normal and COVID-19 pandemic scenarios.

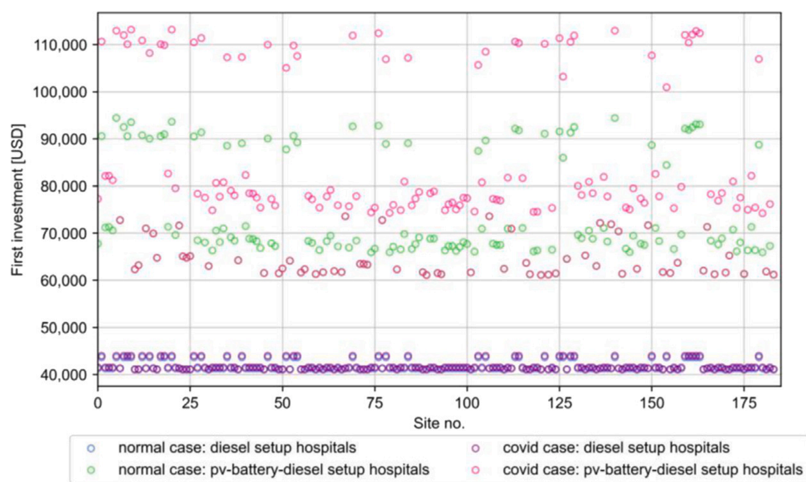


Fig. 11. ESM result: Initial investment for different system setups under normal and COVID-19 pandemic scenarios.

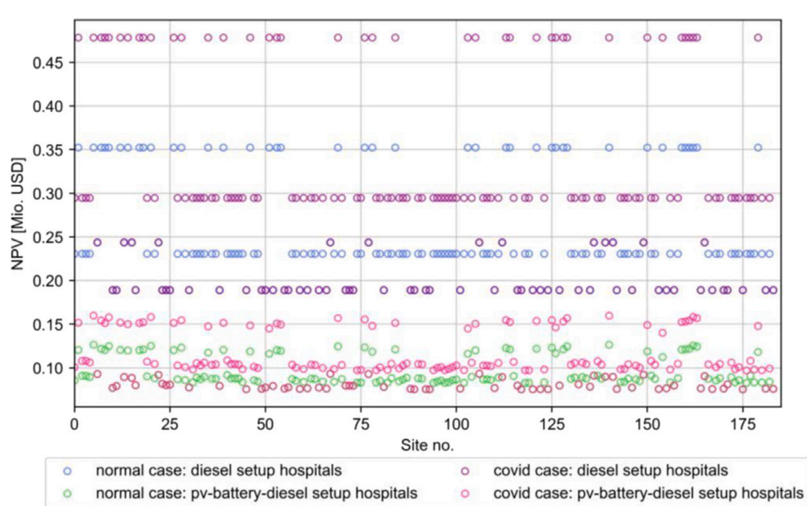


Fig. 12. ESM result: NPV for different system setups under normal situation and COVID-19 pandemic scenarios.

from 61,113 USD to 113,155 USD. The diagram shows that the PV-diesel-battery mini-grids had a significantly higher initial investment cost due to the higher costs of PV and storage technologies when compared to diesel generators.

The Net Present Value (NPV) is a financial metric that considers the initial investment and future costs of operation, maintenance, and replacement over the project's duration. It indicates the cash flow needed to maintain and operate the system. The LCOE is obtained by

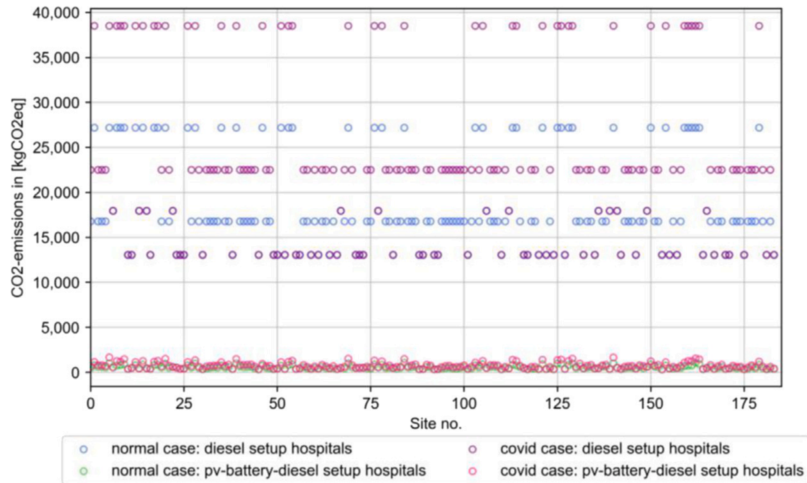


Fig. 13. ESM result: CO_2-emissions for different system setups as well as for normal situation and COVID-19 pandemic situation.

dividing the NPV by the electricity produced by the energy system. Fig. 12 illustrates that the hybrid system setup had a significantly lower NPV than the diesel setup, primarily due to the high diesel fuel cost over the project's duration. This suggests that hybrid system setups are more favourable for the electrification of HCFs.

The CO_2-emissions for all HCF types and scenarios were up to 30 times higher for the diesel setup compared to the hybrid system setup. Additionally, the CO_2-emissions were higher for HCF types with COVID wards during the pandemic due to the extra electricity demand for additional appliances. This is depicted in Fig. 13. These findings align with prior research, which emphasised the significant environmental impact of diesel-only systems in off-grid areas (Antonanzas-Torres, Antonanzas, & Blanco-Fernandez, 2021).

The Offgridders output parameters as presented in Tables 4, 5, 6, and 7 were crucial input criteria for the subsequent MCA to develop recommendations for the prioritisation of electrification for the analysed HCFs.

Multi-criteria analysis

This section presents the results of the two MCAs undertaken for this study, as introduced in Section "Application of Multi-Criteria Analysis to Case Study". Both subjective (expert interviews) and objective (entropy) weighting methods were applied to evaluate their performance on the MCAs' outcomes. To enhance clarity, the following abbreviations define different mini-grid setup options: diesel-only and PV-battery-diesel.

First multi-criteria analysis — evaluating the best energy system setup per healthcare facility site

To prioritise mini-grid electrification of HCFs, the most cost-effective, environmentally friendly, and socially accepted setup per site was determined.

A comparison of results from both weighting schemes is shown in Table 8. Due to the greater variability in the CO_2 *emissions* criterion, the entropy method assigned it a higher weight, while *1st. investment costs* received the lowest objective weighting. Conversely, experts attributed greater importance to *1st. investment costs* than to CO_2 *emissions*, influencing the final MCA results.

Since social acceptance and benefits were beyond the scope of this study, only economic and environmental criteria were considered. Table 9 illustrates the ranking of alternatives for the specific hospital Abromase Health Centre. A similar analysis was conducted for all 184 selected HCFs. The column "Ob. W." represents objective weights, while results using subjective weights are found in Table 10.

Table 8
Comparison between subjective and objective weighting schemes for first MCA.

Weighting scheme	Weights			
	1st inv. costs	O & M	LCOE	CO_2 emissions
Subjective	0.28	0.26	0.26	0.20
Objective	0.05	0.19	0.12	0.64

Using the VIKOR method (see Section "Ranking According to the VIKOR Method"), alternatives were ranked based on Q values. Across both weighting models, the PV-battery-diesel setup consistently ranked highest, followed by the diesel-only systems. Although subjective weighting did not fulfil the "acceptable advantage" condition (see Section "Ranking According to the VIKOR Method"), both models confirmed the superiority of PV-battery-diesel, making diesel-only alternatives less favourable.

Second multi-criteria analysis — prioritising electrification of healthcare facilities

Having established the most suitable energy system setup, the next step was to prioritise HCF electrification using the same methodology. New criteria were introduced, and their weights were calculated. A comparison of both weighting schemes is provided in Table 11.

The prioritisation criteria included:

1. *1st investment costs* — Initial capital required for electrification.
2. *Operational & Maintenance (O&M) costs* — Long-term economic sustainability.
3. *Distance to the grid* — Grid extension feasibility and cost.
4. *Households reached* — Number of people potentially benefiting from electrification.

A key divergence emerged between subjective and objective weighting schemes. The entropy method overwhelmingly prioritised *households reached* (0.76), while economic criteria received near-negligible weights (0.006 and 0.003 for *1st investment costs* and *O&M costs* respectively). In contrast, subjective weighting distributed importance more evenly, giving greater emphasis to financial considerations.

This discrepancy led to contrasting prioritisation outcomes. Objective weighting, ranked facilities serving the largest populations highest, whereas subjective weighting produced a more balanced prioritisation, factoring in cost-effectiveness and accessibility. This highlights a fundamental trade-off: prioritisation based solely on impact (number of beneficiaries) versus incorporating financial and logistical feasibility.

Table 9
Example for abromase health centre.

Category	Criterion	Unit	Ob. W.	Alternatives	
				Diesel-only	PV-hybrid
				W. scores	
Economic	1st inv. costs	US $	**0.05**	0.00	0.03
	O & M costs	US $	**0.19**	0.12	0.001
	LCOE	US $	**0.12**	0.11	0.01
Environ.	CO_2 ems.	kg CO_2 eq	**0.64**	0.39	0.01
S				0.95	0.05
R				0.64	0.05
Q				**1**	**0**
Rank				2	1

Table 10
Example for abromase health centre.

Category	Criterion	Unit	Sj. W.	Alternatives	
				Diesel-only	PV-hybrid
				Values	
Economic	1st inv. costs	US $	**0.28**	0.00	0,14
	O&M costs	US $	**0.26**	0.16	0.002
	LCOE	US $	**0.26**	0.24	0.02
Environ.	CO_2 ems.	kg CO_2 eq	**0.20**	0.12	0.001
S				0.72	0.28
R				0.26	0.28
Q				**0.93**	**0.37**
Rank				2	1

Prioritisation decisions are inherently influenced by value judgments. While objective methods ensure consistency and eliminate bias, they risk overlooking economic feasibility. Conversely, subjective weighting ensures practical implementation but may reduce immediate social impact.

As in the first MCA, ranking was performed using VIKOR (Tables 9 and 10). To maintain readability, full results are available in Appendix “MCA results”, with Tables 12 and 13 presenting a subset. Fig. 14 provides a visual comparison of prioritisation outcomes under subjective (a) and objective (b) weighting.

Unlike the previous MCA, both subjective and objective weighting models met VIKOR’s conditions (Section “Ranking According to the VIKOR Method”), meaning the ranking in Q balanced “group utility” (S) and the minimum individual regret (R) (Opricovic & Tzeng, 2004).

However, ranking discrepancies remained substantial. Only the highest-ranked facility in the objective analysis appeared in the top five under subjective weighting. This confirms the impact of weighting choices on prioritisation results. Expanding the number of criteria, particularly social impact dimensions, could significantly influence these outcomes (see Section on limitations). Additionally, alternative weighting techniques, such as those discussed in Juanpera et al. (2020), Rao, Sai, and Babu (2017) and San Cristóbal (2011), may yield different perspectives. These findings align with previous studies on unreliable electricity supply in rural HCFs. For instance, Lazo et al. observed similar critical power supply challenges during COVID-19 (Lazo et al., 2023). Moreover, our findings extend the work of Aemro, Moura, and de Almeida (2022) by exploring PV hybrid mini-grids as an electrification option for rural HCFs, offering a novel perspective on advising electrification strategies through a MCA.

Ultimately, these results highlight the significant role of methodological choices in determining electrification priorities. Different approaches lead to substantially different yet equally valid rankings, each with real-world implications for healthcare infrastructure development. The decision of which HCF to electrify first should, therefore not be taken lightly, as it impacts both cost efficiency and social equity in energy access.

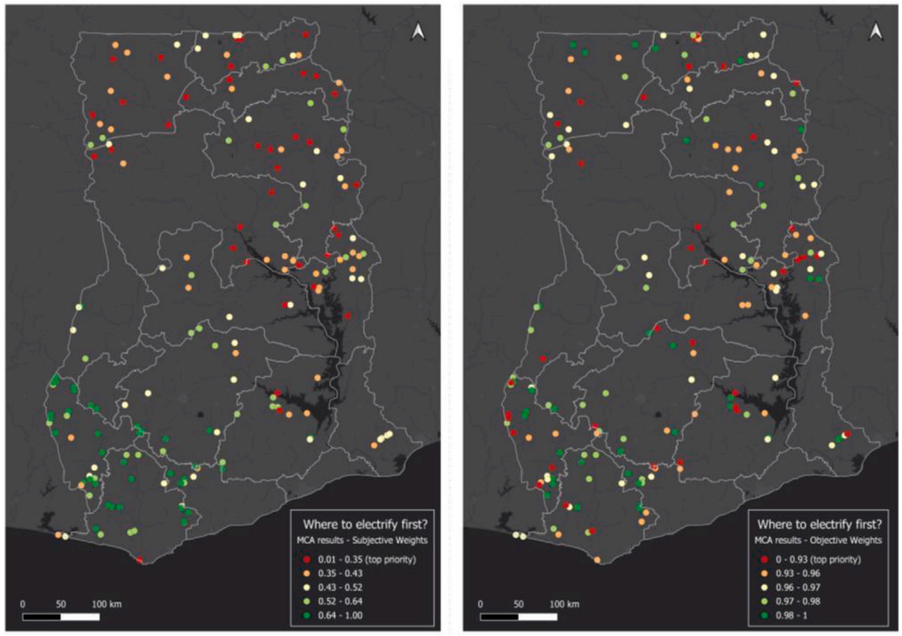
Limitations

There is no universal, step-by-step formula for developing a project like the one presented here. While an integrated methodological approach to HCF electrification provides structure, its implementation in practice requires ongoing decision-making and flexibility in response to unforeseen challenges. This section examines the limitations encountered while following the proposed approach and suggests potential strategies for addressing them in future research.

First, the population data pre-processing in Section “Available Datasets” has a limitation: Values from CIESIN and Facebook datasets might not have been captured accurately when merged with GRID3 settlement polygons. Since these datasets use different methods, full alignment was unlikely, potentially omitting rural populations present in only one source and reducing the accuracy of household coverage estimates (Section “Analysis of population coverage”). Until a unified geo-referenced dataset becomes available, the 20 m buffer method (Section “Population”) remains the best approach to maximise population inclusion. However, cultural and economic biases may be embedded in these datasets, as data collection methods often underrepresent marginalised or nomadic groups, areas with lower connectivity, or regions with limited digital infrastructure. These biases could skew population estimates and impact prioritisation outcomes.

Second, based on the discrepancy between the total number of HCFs estimated by us and the number estimated in the literature, the number of households covered could be much higher with a fully up-to-date and georeferenced dataset on Ghana’s HCFs. However, as established in Section “Healthcare system in Ghana”, not all HCFs types provide the same services, so claiming that a percentage of population is covered based on our buffering is a rough estimation and must be read with care. Moreover, regional variations in healthcare delivery and administrative record-keeping may result in an uneven representation of facilities across regions, particularly disadvantaging under-resourced areas in Northern Ghana. These regional disparities may introduce structural bias in the analysis and should be explicitly accounted for in future studies.

Third, as this paper introduces a new approach to integrated healthcare and electrification planning, certain simplifications were made to reduce complexity and facilitate implementation. However, future research should refine the methodology to better account for the complexities of rural healthcare systems. One key simplification was the assumption that HCF demand remains the same for facilities of the same type, regardless of location. In reality, demand varies due to local conditions and population needs. Additionally, reasonable walking distances were standardised across all regions and population groups, despite factors such as age, mobility limitations, and care-giving responsibilities influencing what is considered an acceptable distance. Cultural norms around healthcare-seeking behaviour, particularly among women and minority communities, as well as differences in perceived quality of care, may further affect utilisation patterns. These aspects should be integrated into future iterations to enhance contextual relevance and equity in planning.



(a) (b)

Fig. 14. Prioritisation results following the VIKOR method using subjective weights (a); Prioritisation results using objective weights (b).

Table 11
Comparison between subjective and objective weighting schemes for second MCA.

Weighting scheme	Weights			
	1st inv. costs	O & M	Distance to grid	Households reached
Subjective	0.28	0.26	0.24	0.21
Objective	0.006	0.003	0.22	0.76

Table 12
Prioritisation for electrification of HCFs and subjective criteria weighting.

Project site name	S	R	Q
Ekye Health Centre	0.406	0.205	0.014
Kpatinga Health Centre	0.480	0.207	0.143
Buma Health Centre	0.409	0.238	0.178
K.N.East Health Centre	0.529	0.202	0.198
Damanko Health Centre	0.519	0.209	0.215
...	...	...	...
Sunkwa Clinic	0.760	0.262	0.857
Amonie Community Clinic	0.763	0.270	0.901
Agona Amenfi Community Clinic	0.769	0.269	0.905
Papueso Presby. Clinic	0.789	0.280	0.990
Adjoum Community Clinic	0.795	0.280	1.000

Table 13
Prioritisation for electrification of HCFs using objective criteria weighting.

Project site name	S	R	Q
Buma Health Centre	0.228	0.225	0.000
Naaga CHPS	0.480	0.254	0.226
Tuluwe HealthCentre	0.517	0.296	0.285
Kabiti CHPS	0.565	0.352	0.363
Bunburika CHPS	0.609	0.381	0.419
...	...	...	...
Alhajikrom PHC Clinic	0.989	0.758	0.992
Mempeasem Health Centre	0.988	0.760	0.992
Ahmadiyya Clinic	0.987	0.762	0.993
Adom Maternity Home	0.989	0.762	0.994
Bukunor Community Clinic	0.995	0.762	0.999

Fourth, several limitations relate to the MCA, which serves as an initial framework for structuring electrification planning in the health sector. The criteria did not account for electricity supply shortages, excluding reliability considerations and potentially reducing the accuracy of decision-making. Future research should integrate energy reliability and demand fulfilment to better assess optimal mini-grid setups for HCFs. Furthermore, economic disparities between regions – such as income levels and funding capacities – were not explicitly included in the prioritisation framework, despite their influence on project feasibility and long-term sustainability.

Lastly, social impact factors – such as user acceptance, affordability, job creation, minority inclusion, and institutional alignment – are crucial but were not included due to data collection challenges, as they typically require qualitative methods like on-the-ground surveys. We acknowledge that our model does not explicitly account for cultural or socio-economic factors, which may influence the outcomes and introduce certain limitations. For instance, cultural resistance to new technologies, traditional healthcare preferences, or distrust in formal institutions may affect uptake and sustainability. While essential for a comprehensive multi-criteria decision analysis, these aspects fell beyond our study's scope. Electrification decisions also have ethical implications, influencing social equity, opportunity access, and long-term sustainability. Prioritisation choices can exacerbate inequalities, and affordability barriers may limit benefits. Ensuring marginalised groups are included in decision-making is key to equitable energy access. Future research should explore these ethical dimensions and explicitly address cultural, economic, and regional biases to develop fair and inclusive electrification frameworks.

Future studies could address these limitations to provide a more comprehensive understanding of their impact on rural electrification and related health outcomes.

Conclusion

Addressed research gap

There is an undeniable link between access to reliable and clean energy and the provision of quality healthcare. Electrification efforts impact healthcare and broader Sustainable Development Goals (SDGs), particularly SDG (Good Health and Well-being) and SDG 7 (Affordable and Clean Energy). Despite tremendous improvements in increasing access to electricity for the Ghanaian population in recent decades, a marked urban–rural divide still leaves thousands of rural settlements without reliable electricity access. As a result, the quality and range of health services in Ghana's HCFs vary by location and access to reliable electricity for powering medical equipment. This exacerbates health inequalities between urban and rural areas, as unreliable energy access disproportionately affects remote communities that already struggle with limited medical resources.

Beyond Ghana, rural electrification remains a major barrier to healthcare delivery in many sub-Saharan African countries. Studies in Nigeria, Ethiopia, and Tanzania have highlighted similar challenges where weak-grid and off-grid HCFs experience service disruptions due to unreliable power supply (Aemro et al., 2022; Khogali et al., 2022; Shastry & Rai, 2021). The proposed hybrid solutions offer a scalable framework for other sub-Saharan African countries with similar energy access challenges. Tailoring this approach to local conditions, including regulatory environments and financial constraints, can ensure effective implementation across different contexts.

In the Ghanaian context, most studies focus on either access to energy or access to quality healthcare. However, little research integrates both aspects within a single framework to inform interventions. This study sought to address this gap by combining geospatial analysis, electricity demand estimation, and a multi-criteria decision approach to guide strategic healthcare electrification efforts. By contextualising energy access as a fundamental determinant of health, this research aligns with global discussions on energy justice and health equity, reinforcing the need for cross-sectoral policies that link energy infrastructure development with healthcare planning. However, limited availability and granularity of HCF data challenge accurate assessment of healthcare demand and energy needs. Future research should address these gaps by incorporating ground-truth data collection and stakeholder engagement to enhance decision-making accuracy.

Summary of findings

With the use of geospatial analysis, we identified that 7% of Ghana's HCFs (184) are located in weak- or off-grid areas. Among them, 172 are public institutions and therefore responsible for vaccine distribution, which is almost impossible without access to electricity to maintain the integrity of the cold chain. In total, these HCFs serve a population of approximately 930,000 people within their catchment area (5 km radius) who would have to drive or walk further to receive vaccinations or more complex medical treatments that require electronic medical equipment. In other words, nearly one million people would benefit from improved healthcare through having reliable electricity to operate refrigerators and electrical medical equipment in Ghana. This underscores the urgent need to integrate electrification efforts into national health strategies to enhance service delivery.

This study also investigated the suitable options to electrify the identified HCFs and analysed the electricity demand profiles of different HCF types, including the impact of additional COVID-19 treatment-related demand surges. The findings revealed that for larger and more complex HCFs, the introduction of specialised COVID-19 wards and equipment led to a demand increase of up to 60%, highlighting the need for scalable and resilient energy solutions.

Based on the results of the first MCA analysing the best option to electrify the analysed HCFs, we found that PV-battery-diesel hybrid system setups are more advantageous than diesel-only systems in both economic and environmental terms. Specifically diesel-only systems generate 30 times higher CO_2 emissions and have two to three times higher LCOE compared to hybrid systems. Moreover, the inclusion of COVID-19 treatment-related electricity demand in system planning was found to reduce electricity costs by up to 0.04 USD/kWh, reinforcing the economic viability of hybrid energy solutions.

To further refine electrification strategies, a second MCA was conducted, incorporating key criteria such as initial investment costs, operating expenses, grid distance, and population coverage. This provided a structured approach to prioritising HCF electrification based on impact and feasibility.

This study contributes to the growing body of research on sustainable healthcare electrification in rural Africa. Our results align with previous work, confirming the economic and environmental benefits of PV-hybrid mini-grids over diesel-only solutions. Additionally, our electrification prioritisation methodology builds upon existing MCA approaches, offering a practical framework for decision-makers. To further validate these findings, pilot projects should be initiated, allowing health-specific real-world testing of hybrid systems in selected HCFs.

Recommendations and outlook

Practical implications and advancing implementation: The implementation of hybrid RE systems – particularly PV-battery-diesel configurations – is a promising solution to reliably meet the demand for HCFs in remote areas. As shown, they have the advantage of being more economically viable, as well as reducing CO_2 and noise emissions, compared to commonly used diesel-powered solutions. To facilitate implementation, policymakers should consider a phased deployment strategy, for example over a five-year period, to streamline electrification efforts in the health sector. This strategy could be oriented along the findings of this study by starting with the high-priority HCFs identified in the study. These efforts should be accompanied by enabling policies that combine strategic development in the health and energy sectors, viewing them as a nexus rather than planning them separately. As mentioned at the beginning of this study, improving access to sustainable electricity, particularly in remote areas, can help attract and retain qualified workers, thereby improving access to quality healthcare and reducing migration to cities. Implementation always requires funding. Sources of funding could include public–private partnerships, international development grants and green energy financing mechanisms. Collaborating with local utilities and technology providers is crucial for ensuring long-term sustainability.

Future research: Looking ahead, future research could benefit from several methodological enhancements to build on the current pre-feasibility analysis. These include incorporating more advanced statistical techniques – such as confidence interval estimation, model diagnostics (e.g., multicollinearity and residual analysis), sensitivity analysis, and external validation – to strengthen the robustness of the findings.

It should be noted that the electrification of HCFs poses the challenge of irregular electricity demand throughout the day, for example due to the use of high demand appliances at certain times. In order to balance these peaks in demand, future work could consider including the settlements selected here for mini-grid electrification, so that not only the HCFs, but also the settlements in their immediate vicinity are supplied with electricity autonomously. Household demand could have a more stable demand profile compared to HCFs, increasing the load factor, and having a positive impact on energy system design and cost distribution. In this way, efforts would be made to achieve healthy, resilient communities with access to both quality healthcare and sustainable, reliable electricity.

Additionally, expanding the interdisciplinary scope to include expertise in socio-economic modelling and public policy could allow for a more comprehensive evaluation of the broader impacts of rural electrification, including migration and health-related outcomes. Such developments would support more holistic and actionable policy recommendations.

In summary, the approach developed proved to be an effective measure of identifying HCFs located in off-grid or weak-grid areas, determining their population coverage, assessing their most appropriate electrification option, and developing recommendations for streamlining electrification planning at the intersection of the health and energy sectors. By adapting this framework to other sub-Saharan African countries considering the above recommendations, stakeholders can apply it to address region-specific constraints while ensuring equitable access to healthcare.

CRediT authorship contribution statement

Andrés Andrade Velásquez: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Abdulrahman Alsanad:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Katrin Lammers:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Avia Linke:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Nathan Johnson:** Writing – review & editing, Supervision. **Elena van Hove:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Catherina Cader:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The research has been conducted within the framework of the EnerShelF project—BMBF funding Grant Number 03SF0567C.

The VNP46A3/VJ146A3 Monthly and VNP46A4/VJ146A4 Yearly moonlight-adjusted Nighttime Lights (NTL) were acquired from Level-1 and Atmosphere Archive & Distribution System (LAADS) Distributed Active Archive Center (DAAC), located in Goddard Space Flight Center, Greenbelt, Maryland.[1]

Appendix A

To provide access to all the datasets used and the results produced, we used the Harvard Dataverse, which enables large datasets and information to be shared quickly and easily. Below you will find links to downloadable files related to our research outlined in this paper.

A.1 Demand modelling

The necessary demand profiles of analysed HCFs were determined using the stochastic modelling tool RAMP. The input tables for running the RAMP tool and the corresponding output files (load profiles) can be found using the following link:

https://doi.org/10.7910/DVN/4GAOGZ

[1] https://ladsweb.nascom.nasa.gov/.

A.2 energy system modelling

The input tables for running the Offgridders tool including all sources and the corresponding output files (optimised energy systems and sensitivity cases for all identified sites) can be found using the following links:

Input data (techno-economic, scenario definition and simulation settings):

https://doi.org/10.7910/DVN/JDALVZ

Input time-series:

https://doi.org/10.7910/DVN/D9LINN

Energy system modelling results:

https://doi.org/10.7910/DVN/T6F42C

A.3 MCA questionnaire

The questionnaire used to ask the experts at the Reiner Lemoine Institut for their assessment of the most important criteria when choosing between different options for setting up a mini-grid system for a single site, and then prioritising the sites to be electrified, is available as a PDF document at the following link:

https://doi.org/10.7910/DVN/YHM8SB

A.4 MCA results

The resulting tables from both MCAs are available as CSV files at the following link:

https://doi.org/10.7910/DVN/T4XOZC

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